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1 Introduction

This section will provide a brief overview on the...It will also provide an overview on the project, it's objectives...

1.1 Project Background

Social media, sometimes referred to as Social Networking Sites (SNS), have been adopted as an important piece of daily life for a substantial number of people in the world. They have provided us the ability to connect with groups of individuals online from around the globe on a never-before-seen scale, allowing for friendships, communities and relationships to be built from virtually anywhere. Dating back to the days of MySpace in the mid-2000s, where at one point, it made up nearly 75% of all social networking traffic and was the most visited website in the United States (Barnett 2024), to social media networks such as X (formerly known as Twitter) and Facebook reaching out to billions around the globe, to the ever-growing trend of short-form-video social networks kickstarted by TikTok, forcing networks such as Instagram and YouTube to adapt to the new medium, social media looks to be omnipresent within a majority of peoples' lives, with it being estimated that 1 in 3 people globally have some form of social media presence (Shah 2021).

Similarly, smartphones have equally taken the world by storm, and have also become an important part of life for most people, even at early ages, with studies reporting that young people are receiving their first phone at the age of 10 (Chan, Yeo, and Handayani 2023). With the June 2007 release of Apple's "iPhone" and the release of the HTC Dream running the open source "Android" operating system in 2008, both Apple's iPhone and the myriad of devices that use Google's Android OS have cultivated a large shift in the world of online media (Goggin 2012), with it being reported in 2021 that 85% of adults in the United States own a smartphone (Ouellette and Rao 2022). Smartphones nowadays are used for communication via calling or texting through apps such as WhatsApp, for entertainment (gaming, live sports, TV, online content, etc..), for productivity, for learning, and a wide variety of other purposes, all contained within a device typically the size of someone's hand. It cannot be understated how much of an overall game-changer smartphones have been.

As time went on though, it was only a matter of time before social media was integrated onto smartphones via downloadable apps from various app storefronts such as Apple's App Store and Google's Google Play, social media networks such as YouTube, X, and Facebook adapted their web interfaces to smartphones, and have been rewarded with large gains in userbase for all, with it being revealed that 80% of content consumed on social media is through a smartphone (Islam et al.

2025). However, over the years, worries have been raised regarding a variety of issues that concern both smartphones and social media, particularly surrounding the problem of individuals developing an overdependence on their smartphones and / or social media that negatively affects their day-to-day life. Commonly categorised as "Smartphone Addiction" and "Social Media Addiction", the issue has become a largely discussed and researched topic over the last 15 years with researchers looking into potential causes and correlations with other health effects.

New forms of media "harming the minds of the youth" isn't a new concept by any means. Going as far back to the invention of the radio, worries from parents about addictions related to radio, television, video games, the internet, and smartphones affecting young people have been prevalent, due to the youth's carelessness in regards to new technologies, but also due to the dystopian over-exaggeration on the dangerous effects that these at-the-time new technologies could have on society (Abeele and Mohr 2021; Adorjan and Ricciardelli 2021). This isn't to say their concerns aren't valid, but rather to point out that the sentiment has been shared with emerging digital mediums.

While Smartphone Addiction isn't classified the same way drug addiction is classified, the nature of it is instead classified as a behavioral addiction (King and Dong 2017), and a similar manner, Social Media Addiction is considered as a subset of Internet Addiction (Hou et al. 2019), which itself is considered a subset of behavioural addictions (Chou, Condron, and Belland 2005), and while both are typically categorised as separate types of "addiction", it has been argued by some that smartphone addiction may be correlated with Social Media Addiction, due to smartphones themselves being enablers for addictive behaviors, including social media addiction (Kuss and Griffiths 2017) (self cite also?). Despite this, the use of the term "addiction" has had pushback, due to the sensationalist language the term brings (Adorjan and Ricciardelli 2021), and also because both terms don't fit the typical perception of addiction with the term "Problematic Smartphone/Social Media Use" being used as an alternative term (Rozgonjuk, Saal, and Täht 2018).

The nature of Smartphone Addiction and Social Media Addiction has on young individuals have been extensively studied over the years, including why individuals may develop an addiction in the first place. When looking at why some may be addicted to smartphones and social media, a study from Indian researchers, Bhalerao et al. (2020), that analysed smartphone addiction provided a multitude of possible causes behind Smartphone Addiction, one of which being how the current generation has been brought up alongside the internet and technology as a whole, therefore they may feel a preference to communicate over the internet as opposed to in-person, fu-

eling the dwindling need for talking face-to-face with someone. Another study from Turkish researchers (Özcan and Ozer 2017) mentioned how alongside young people (defined as between 18-29) ranking first in internet usage, they're also more prone to problematic smartphone use, possibly due to the characteristics the phone carries that triggers heavy smartphone usage. A 2018 study from Turkish researcher Aksoy (2018) looked to define a set of causes behind social media addiction and found that most of those who considered themselves addicted to social media were addicted due to factors such as a lack of friends outside of social media, the fact that social media was a "social activity requirement", a need to fulfil a task of some kind, ensuring that they were kept updated on current events, and the fact that social media was now ingrained with social life in a general sense.

Many studies have also looked at the multitude of effects that are correlated with both Smartphone and Social Media Addiction, for example, in 2017, researchers in the United States King and Dong (2017) made a series of hypotheses in relation to college students and the effects related to smartphone addiction, in particular, they looked at smartphone addiction's associations with "life stressors" (a measure of stress and anxiety), "digital overload" (a measure of information and communication overload), retention, academic performance, and social well-being. The results of their study found that smartphone addiction could be positively correlated with life stressors, digital overload, retention, and academic performance, with social well-being as the only factor with no significant correlation. Following the results, they concluded by mentioning that 3 out of their 5 hypotheses were consistent with past studies, while mentioning that regarding the hypothesis related to retention was "...a novel insight created from this study as current smartphone addiction has not linked the consideration with college retention.". Another study from researchers Lin et al. (2016) looked at the correlation between depression and social media use in young American adults, and found that those who exhibited heavier social media use tended to show a high indication of depression.

That being said, many researchers and developers over the years have made attempts to study and provide treatment for the issue of Smartphone and Social Media Addiction, both as experimental studies on alternative treatments and as dedicated mobile applications. A 2024 India-based study (Seran 2024) aimed to look at using educational-based methods to reduce social media addiction and obesity. After answering a questionnaire, participants were split into a control group and a experimental group, the latter of which were then involved in educational activities such as lectures, brainstorming sessions, and exercises, among other activities. Regarding the results for social media addiction, participants were then scored on their social media addiction one, three, and six months after the initial study. The researchers

found that there was a 21.2% difference in social media addiction scores between the control group and the group that partook in the educational-based intervention, with them concluding that it was an effective measure in reducing social media addiction.

For specifically mobile applications, in 2013, a group of German researchers Löchtefeld, Böhmer, and Ganev (2013), developed an android-based mobile application called "AppDetox" an app that's main objective was to provide a "medication" for smartphone addiction, with the app allowing users to set restriction-based rules on apps that prevent the user from opening the app at specific times of the day, until the user removed the rule, and after a set amount of time. It also displayed information on their set rules, times they've broken said rules, and usage statistics on all their apps. Their analysis found that users mostly set restrictions on messaging apps such as WhatsApp and WeChat, and broke self-created rules at roughly a 7:1 ratio (7 times a rules was broken for every 1 rule set).

However, there seemingly exists a gap of intervention apps that deal specifically with smartphone or social media addiction, and that also educate the user. Most literature that's within the topic of addiction provides analysis on apps that deal with adjacent issues. For example, a 2021 study (Oliveira et al. 2021) reviewed the effectiveness of psychological intervention mobile apps to help with issues such as depression, anxiety, and a user's overall mental health, and while they concluded that the apps in the selected literature showed promise in reducing the aforementioned issues, none of the literature reviewed looked at specifically treating the issue of smartphone or social media addiction.

1.2 Project Overview

1.2.1 Project Outline

(something about providing a solution while also implementing an experimental method, in this case, reality therapy)

While evidence has shown psychology-based approaches in intervention mobile applications to be promising in solving the issues of smartphone and social media addiction, very few apps have been developed and released with this approach in mind, with most consumer products on the app store opting to utilise restriction and time-out technologies instead. Because of this approach, young adults aren't given much opportunity to reflect on their addiction and the root cause of it should they choose not to seek outside help, leading to very little growth for the end-user. However, by taking this psychology-based approach, young adults can be given opportunity to reflect on their addiction and learn how to treat the issue. **(PLACEHOLDER,**

REWORD AND SELF-CITE)

To create a mobile application that integrates reality therapy to help treat smartphone and social media addiction

1.3 Project Objectives

1.3.1 Secondary Objectives

Investigate how mobile intervention apps attempt to encourage outside activity from its users
Research how past and present mobile intervention applications encourage its users to engage in other activities independent from the smartphone. It will also involve looking at common suggestions of outside activities to suggest to users to replace their smartphone usage with.

Investigate the use of methods based in psychology used to treat smartphone and social media addiction
Research how psychological methods have been used in the past to treat smartphone and social media addiction, and see if such methods have been integrated with mobile apps to gain a further understanding on how these methods could be integrated or improved upon.

Investigate how intervention apps ensure users stay on track with their treatment over long periods of time
Research how existing intervention apps try to engage and encourage its users and ensure they keep on track with maintaining reduced smartphone usage. This will give a good sense on how to make sure potential end-users stay engaged with the app and the treatment as a whole. (expand)

Investigate how intervention apps contextualise smartphone addiction to educate its users while handling the topic gracefully
Research how (or if) intervention apps have tried to communicate the dangers of smartphone addiction and why it's important to maintain healthy smartphone usage, while handling it in a graceful manner. This will help in finding the kind of tone the app should take when providing information to the user, encouraging them to take the "treatment" seriously. It will also suggest if there is an educational gap in these types of apps.

1.3.2 Primary Objectives

Develop the smartphone intervention app
Create the application based on the knowledge gained findings from the literature and technology review.

Another primary objective (temp)
text goes here

some kind of testing (temp)

some kind of analysis and conclude (temp)

2 Literature and Technology Review

2.1 Investigate how mobile intervention apps attempt to encourage outside activity from its users

2.2 Investigate the use of methods based in psychology used to treat smartphone and social media addiction

2.3 Investigate how intervention apps ensure users stay on track with their treatment over long periods of time

2.4 Investigate how intervention apps contextualise smartphone addiction to educate its users while handling the topic gracefully

3 Project's Primary Research Methods

This section will give a detailed overview on the methods that will be used to carry out the remainder of the project and provide a reasonable conclusion to the problem statement. This section will also outline what work has been completed up to this point and all remaining work that will be carried out over the project's lifetime.

3.1 Research Method and Justification

3.2 Develop the Mobile app (temp)

3.2.1 Development

3.2.2 Project Lifecycle

3.3 Usability Testing with users and Evaluation

3.4 Preliminary Work

3.5 Remaining Work

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